

Divide $5x^3 + x^2 + 2x + 6$ by $3x^2 + 1$ in $\mathbb{Z}_7[x]$.

"4x

"5

$$\left(\frac{5}{3}x \bmod 7\right) + \left(\frac{1}{3} \bmod 7\right)$$

$$3x^2 + 1 \overline{) 5x^3 + x^2 + 2x + 6}$$

$$-(5x^3 + 0x^2 + 4x)$$

$$1 \cdot x^2 + 5x + 6$$

$$-(1 \cdot x^2 + 0x + 5)$$

$$5x + 1 = \text{REM.}$$

Therefore

$$\text{quo}(5x^3 + x^2 + 2x + 6, 3x^2 + 1)$$

$$\equiv 4x + 5 \bmod 7$$

$$\text{rem}(5x^3 + x^2 + 2x + 6, 3x^2 + 1)$$

$$\equiv 5x + 1 \bmod 7.$$